Infosys_HotelRevAI – AI Driven Revenue Analysis for Hotels

Milestone 2: Data Loading & Data Modeling

Progress Report – Milestone 2

Name of the Mentor: Selciya

Team C

Team Members:

1. **Rinisha Shajitha**
2. **Om Bokare**
3. **Rinal Dave**
4. **Mihira Sri**
5. **Aditi**
6. **Alekhya**
7. **Rethiga**

1. Introduction

Data modeling plays a crucial role in transforming raw data into meaningful and structured information that can be used for analysis and decision-making. In business intelligence tools like Power BI, a well-designed data model improves performance, ensures accurate calculations, and makes report creation more efficient.

This documentation describes the data modeling process for the **Hotel Booking dataset**. The objective of this model is to organize booking-related data into a structured format that supports reporting and analytical insights.

The model has been designed using a **Star Schema architecture**, which is a widely accepted best practice in dimensional modeling. This structure separates transactional data from descriptive data, allowing better performance and simplified relationships.

2. Dataset Overview

The dataset used in this project contains hotel booking records, including details about customers, booking dates, hotels, market segments, and countries. The data was structured into fact and dimension tables to support analytical reporting.

The dataset includes the following key information:

- Booking arrival dates
- Hotel names
- Customer types

- Market segments
- Country information
- Other booking-related attributes

Before modeling, the data was cleaned and structured appropriately to ensure consistency and reliability.

3. Data Model Architecture

The data model follows a **Star Schema design**, where a central fact table is surrounded by multiple dimension tables. This structure simplifies querying and improves report performance.

In this model:

- The **Fact_HotelBookings** table acts as the central table.
- Dimension tables provide descriptive attributes for filtering and grouping.

The tables included in the model are:

- **Fact_HotelBookings**
- **DateTable**
- **Dim_Country**
- **Dim_CustomerType**
- **Dim_Hotel**
- **Dim_MarketSegment**

Each dimension table is connected directly to the fact table through a one-to-many relationship

DateTable

|

Dim_Hotel — Fact_HotelBookings — Dim_CustomerType

|

Dim_MarketSegment

|

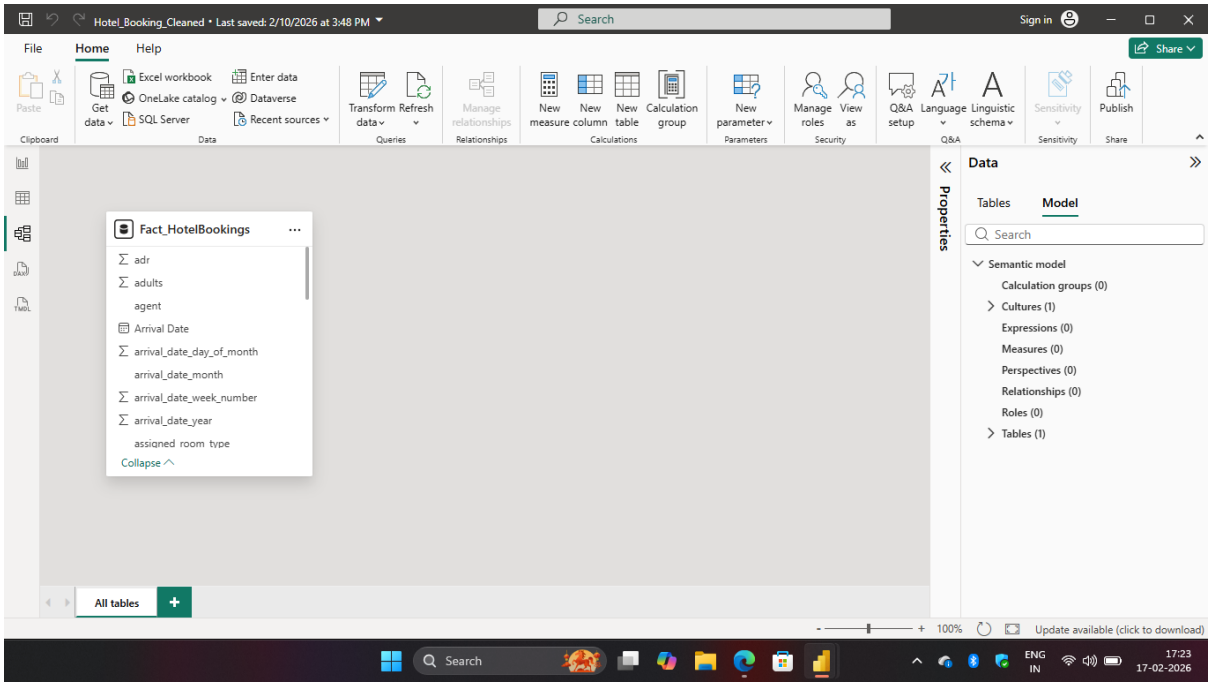
Dim_Country

4. Fact Table Description

The **Fact_HotelBookings** table contains the transactional data for hotel bookings. It serves as the core of the data model and stores measurable and analytical information.

This table includes:

- Arrival Date
- Country
- Hotel
- Customer Type
- Market Segment
- Booking-related metrics



Each record in this table represents an individual booking entry.

The fact table connects to dimension tables using foreign keys, allowing detailed analysis across multiple perspectives such as time, geography, and customer category.

5. Dimension Tables Description

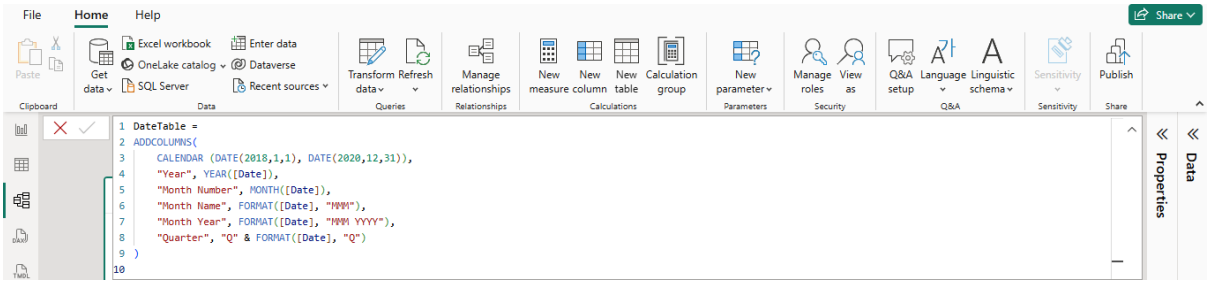
Dimension tables provide descriptive information that helps categorize and filter fact data. They contain unique values and are placed on the "one" side of relationships.

5.1 DateTable

The DateTable is used for time-based analysis and reporting. It enables grouping data by year, quarter, month, and other time attributes.

It includes:

- Date
- Month Name
- Month Number
- Month Year
- Quarter

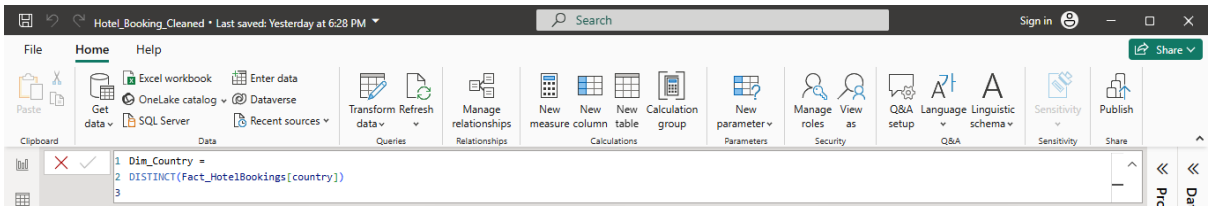


5.2 Dim_Country

The Dim_Country table stores unique country values used in booking records.

This table helps:

- Analyze bookings by country
- Compare regional performance
- Filter reports by geographic location

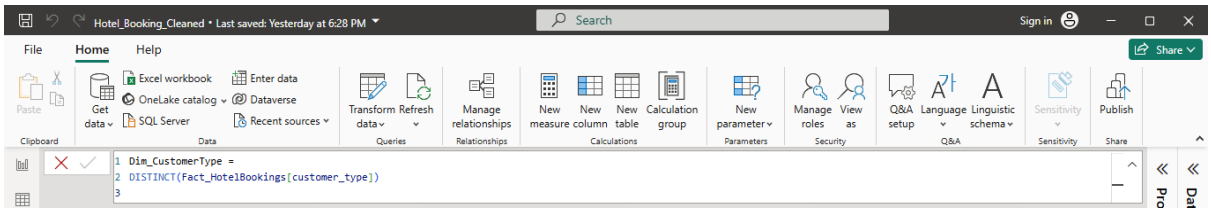


5.3 Dim_CustomerType

This table contains different categories of customers.

It helps in:

- Segmenting bookings by customer type
- Understanding customer behavior
- Comparing booking patterns

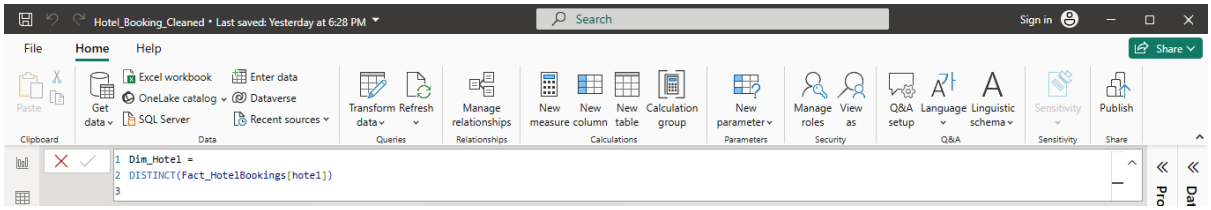


5.4 Dim_Hotel

The Dim_Hotel table includes unique hotel names.

It allows:

- Performance comparison between hotels
- Hotel-level filtering
- Revenue or booking distribution analysis



5.5 Dim_MarketSegment

This table stores market segmentation information.

It helps:

- Analyze booking channels
- Understand target markets
- Compare segment performance

6. Table Relationships

Relationships define how tables interact with each other. In this model, each dimension table is connected to the Fact_HotelBookings table using a one-to-many relationship.

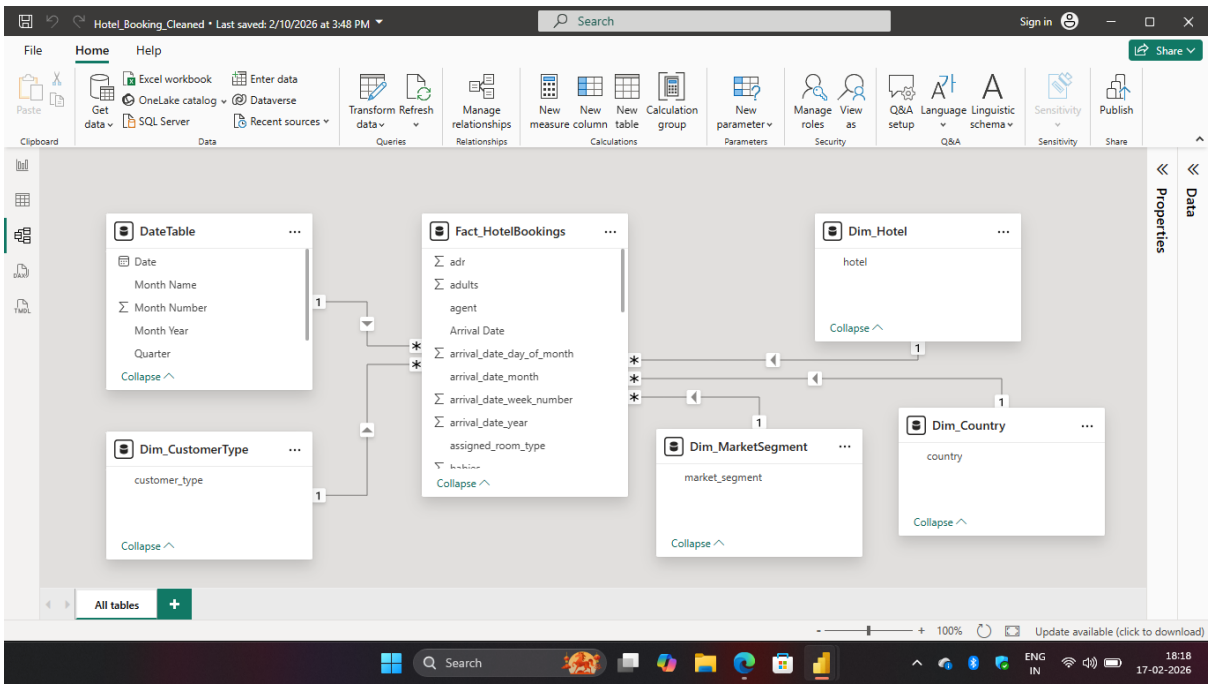
The relationships are defined as follows:

- DateTable[Date] → Fact_HotelBookings[Arrival Date]
- Dim_Country[Country] → Fact_HotelBookings[Country]
- Dim_CustomerType[Customer Type] → Fact_HotelBookings[Customer Type]
- Dim_Hotel[Hotel] → Fact_HotelBookings[Hotel]
- Dim_MarketSegment[Market Segment] → Fact_HotelBookings[Market Segment]

All relationships are:

- Active
- One-to-Many (1 → *)
- Single-direction filtering

Single-direction filtering ensures that dimension tables filter the fact table without causing ambiguity.



7. Cardinality and Filter Direction

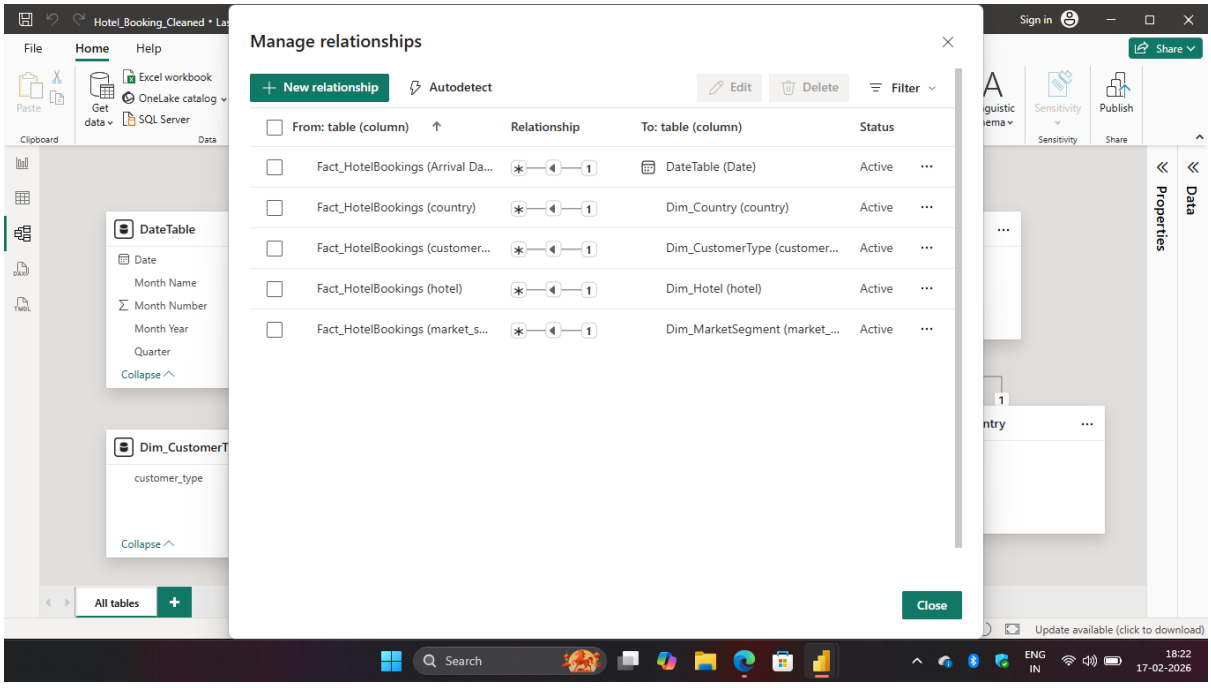
Cardinality defines how tables are related. In this model:

- Dimension tables represent the "One" side
- Fact table represents the "Many" side

This setup ensures:

- Data integrity
- Accurate aggregations
- Better performance

Filter direction is set to single direction from dimension to fact table. This prevents circular dependencies and ambiguous filtering behavior.



8. Benefits of the Data Model

The implemented Star Schema provides several advantages:

- Improved query performance
- Simplified report development
- Reduced redundancy
- Clear separation of transactional and descriptive data
- Better scalability

9. Conclusion

The Hotel Booking data model has been successfully designed using a Star Schema architecture. The separation of fact and dimension tables ensures clarity, performance efficiency, and analytical flexibility.

By implementing proper one-to-many relationships and maintaining single-direction filtering, the model supports reliable reporting and scalable dashboard development.

This structured approach makes the dataset suitable for advanced business intelligence analysis and future expansion.

Overall, the data modeling process has created a strong foundation for generating meaningful insights from hotel booking data.